Module 3
Business Economics

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Output decision
Perfect competition: P = MC => profit-maximising q*.

where P = price
MR = marginal revenue
MC = marginal cost

Monopoly: MR = MC => profit-maximising Q*

where Q = quantity/output

Pricing decision
Perfect competition: Take P as given which in turn is determined by market forces (i.e. market demand and supply of the product).

Monopoly: Given Q*, find the price along the demand curve that maximises the difference between the revenues and costs.

Long-term profitability
Perfect competition: Free entry and exit implies that only zero economic profit is earned in the long run.

Monopoly: Barriers to entry implies the monopoly firm can earn above-normal profit in the long run.

Answer 1(b)(i)

Quantity (unit)	Price (HK$)	Marginal revenue (HK$)	Marginal cost (HK$)
1	23	23	19
2	21	19	19
3	19	15	19
4	17	11	19

Answer 1(b)(ii)

To maximize profit, the condition should be MR(Q) = MC(Q).

At Q = 2 (refer to the table above), MR(Q=2) = MC(Q=2) = HK$19. It follows that the monopoly's profit-maximizing output should be 2 units.

Answer 1(b)(iii)

Profit = (HK$21 x 2) – (HK$19 x 2) = HK$4

Answer 1(b)(iv)

Consumer surplus, which measures consumer benefits, is represented by the difference between the price that consumers pay and the price that they are willing to pay.

Answer 1(b)(v)

Perfect competition: P = MC = HK$19

Size of consumer surplus transferred to the producer (firm)
= (HK$21 – HK$19) x 2
= HK$4

Answer 2(a)

Nominal GDP values current output using current prices, which is not corrected for inflation.

Real GDP values current output with base-year prices, through which inflation is corrected/ removed.

Answer 2(b)(i)

- Government purchases rise by HK$50 million.
- Net exports fall by HK$50 million (as imports increase by HK$50 million).
- As a result, GDP remains unchanged.

Answer 2(b)(ii)

- Consumption rises by HK$140 million.
- Inventory investment rises by HK$60 million.
- GDP rises by HK$200 million.

Answer 2(c)(i)

A fall in the size of labour.

AS curve shifts to the left due to a decrease in the supply of labour, resulting in an increase in the price level and a decrease in the output level (i.e. stagflation).

Answer 2(c)(ii)

An appreciation of the domestic currency.

AD curve shifts to the left due to decrease in net export (the appreciation of the domestic currency against the foreign currencies in general has made the imported foreign goods cheaper in domestic currency while the exported domestic goods more expensive in foreign currency), resulting in a decrease in both price and output level.

Answer 3(a)

Opportunity cost refers to the value of the next highest-valued alternative use of the resources.

Answer 3(b)

Opportunity cost = Direct and indirect costs of the decision
= tuition + books
= HK$50,000 + HK$1,000 = HK$51,000

Answer 4(a)(i)

You should be using arithmetic return to estimate next year (2020)'s yearly return.

Answer 4(a)(ii)

Arithmetic Average Return (Portfolio 1) $= \frac{\sum x_i}{n} = \frac{3 + 35 + 30 + 27 - 15}{5} = 16\%$

Arithmetic Average Return (Portfolio 2) $= \frac{\sum x_i}{n} = \frac{21 + 26 + 18 - 5 + 45}{5} = 21\%$

Answer 4(b)(i)

Range (A) = 35% – (–15%) = 50%

Range (B) = 45% – (–5%) = 50%

Answer 4(b)(ii)

The advantages of the range are:
- It is easy to understand.
- It is easy to calculate.

The disadvantages of the range are:
- It is influenced by extremely high or low values (i.e. outliers).
- It is not representative (of the sample) because only two values are extracted for use from the distribution (i.e., do not use all the observations in the data set).

[Other valid answers are acceptable]

Answer 4(c)(i)

Sample standard deviation

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i-1}^{n}(x_i - \mu)^2}$$

σ (Portfolio 1) = 21.26%

σ (Portfolio 2) = 17.93%

Answer 4(c)(ii)

Standard deviation, which estimates the average distance between each observation and the mean, avoids the outlier and insufficient representativeness (of the sample) problems, as mentioned in part (b).

Answer 4(d)(i)

CV (Portfolio 1) $[= \frac{\sigma_A}{\mu_A} \times 100\%] = \frac{21.26}{16} \times 100\% = 132.88\%$

CV (Portfolio 2) $[= \frac{\sigma_B}{\mu_B} \times 100\%] = \frac{17.93}{21} \times 100\% = 85.38\%$

Answer 4(d)(ii)

Standard deviation could be difficult to interpret when two data sets have markedly different means and/or different units of measurement. This is because standard deviation is unit-sensitive.

Coefficient of variation avoids these problems as it is expressed as a ratio between the standard deviation of the set of data to their mean value and thus is unit-free.

Answer 5(a)

Probability sampling refers to the procedure in which each unit of a population has a known, positive probability of being selected.

Non-probability sampling refers to the procedure in which some elements of the population have no chance of being selected (i.e., have been excluded by the researcher).

Answer 5(b)

Non-probability sampling methods:

1. Quota sampling: The population is first segmented into mutually exclusive subgroups and then judgement is used to select the subjects or units from each segment based on certain criteria.
2. Convenience sampling: The sample is chosen based on access or convenience.
3. Judgement sampling: The sample is chosen based on the researcher's judgement about the subjects to be included.
4. Snowball sampling/chain-referral sampling: The sample is assembled through the recruitment of new subjects with similar traits by existing subjects.
5. Consecutive sampling: The sample is formed by keeping recruiting subjects that meet given criteria until the required sample size is achieved.

[Other valid answers are acceptable.]

* * * END OF EXAMINATION PAPER * * *